VERA: Reliability-Gated Multimodal Anomaly Fusion under Domain Stress

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***Abstract*—This manuscript presents VERA, a reliability-aware fusion pipeline for heterogeneous anomaly detection. The system combines domain experts for fraud, cyber telemetry, user behavior, and text evidence, then learns a fusion layer that produces a unified anomaly risk score and a domain-level explanation. The central research question is whether a practical fusion layer can preserve clean benchmark accuracy while using calibration and drift evidence to reduce brittle behavior under missing, shifted, or corrupted domains.**

The technical contribution is Reliability-Gated Attention (RGA), a reliability-aware fusion module inside VERA. RGA couples masked cross-domain attention with a post-hoc reliability estimator that combines validation expected calibration error, test-time score-distribution drift, and prediction sharpness. A conservative gate preserves the static attention path when domains appear reliable and injects reliability weights only when batch evidence indicates degraded domain quality. The system also includes a reproducible input schema, strong score-level baselines, missing-domain ablations, adversarial perturbation stress tests, calibration analysis, and counterfactual domain attribution.

This version adds a naturally paired benchmark path based on MVTec 3D-AD: paired RGB and 3D/depth observations are converted into the same long-format fusion schema used by the rest of VERA. A local bagel-subset smoke run contains 78 paired observations. In that paired run, random forest fusion and early-fusion MLP are strong, while static attention and RGA are not competitive; this is reported as end-to-end paired-pipeline evidence, not as a performance win. The main local empirical study still reports RealFusion-LA from four real local datasets: Credit Card Fraud, UNSW-NB15, Online Shoppers Intention, and labeled news text. Because those datasets are not naturally co-observed, they are treated as a secondary label-aligned stress benchmark. Across five clean seeds, RGA and static attention are effectively indistinguishable on ROC-AUC and PR-AUC because the label-aligned split is near-saturated. The useful signal is therefore not the saturated clean table but the stress setting: under all-domain zero and max score-collapse attacks, RGA improves ROC-AUC by 0.064 and 0.030 over static attention in the current result artifact. The central claim is conditional: *RGA improves reliability-stress robustness when paired evidence streams undergo coherent confidence collapse, while preserving competitive clean ranking performance.* The manuscript reports that claim with confidence intervals and explicit failure cases rather than as a blanket superiority statement.

***Index Terms*—anomaly detection, multimodal fusion, reliability, calibration, attention, score-level fusion, counterfactual explanation, fraud detection, cybersecurity**

I. Introduction

Operational anomaly detection is usually a system problem before it is a single-model problem. A high-risk case may appear weakly in transaction records, network telemetry, user behavior, text evidence, device signals, or document checks. Each domain expert may be accurate inside its own data distribution, but the final decision still depends on a fusion layer that decides how much to trust each signal. This trust problem becomes harder when domains are missing, score distributions drift, calibration degrades, or an adversary corrupts one or more sources.

VERA is a research system for this setting. The system is not designed as a new fraud detector, network detector, or text detector in isolation. Instead, it treats anomaly detection as a verification pipeline: domain experts generate scores and compact evidence, those scores are converted into a unified fusion schema, and a reliability-aware fusion module produces a final risk estimate with a domain-level explanation. This architecture matches how many practical anomaly systems are deployed: detectors are built and maintained by different teams, and the fusion layer must remain robust as their data quality changes.

The paper treats VERA as the system boundary and RGA as one mechanism inside that boundary. This matters because the contribution is not a new domain detector. It is a reliability-aware fusion layer with explicit input schema, stress tests, and limitations.

A. Research Question

The paper asks:

> *Can a multimodal anomaly-verification system preserve clean score-fusion accuracy while using calibration and drift evidence to reduce brittle behavior under domain failure?*

This question is intentionally narrower than a claim of general anomaly detection superiority. The present codebase already supports domain experts, fusion schemas, attention fusion, reliability scoring, and robustness tests. The research target is the fusion and verification layer that combines those experts under imperfect operating conditions.

B. Thesis and Evidence Standard

The working thesis is that reliability-aware fusion is most useful when the operating condition changes. A strong paper must therefore show three things:

1) clean performance remains competitive with strong baselines,
2) reliability adaptation improves behavior under meaningful stress, and

TABLE I
SYSTEM REQUIREMENTS FOR VERA AND HOW THE CURRENT IMPLEMENTATION ADDRESSES THEM.

Requirement	Current implementation
Heterogeneous evidence	Fraud, cyber, behavior, and text domains are mapped to a shared score-level schema.
Missing-domain support	The attention model uses explicit masks; baselines receive missingness-aware fallbacks.
Reliability awareness	Domain weights combine calibration error, KS drift evidence, and score sharpness.
Strong baselines	Random forest, early fusion MLP, late fusion ensemble, static attention, and confidence weighting are included.
Explanation	Counterfactual domain attribution masks one domain at a time and measures risk change.
Reproducibility	Data generation, benchmark execution, asset generation, and PDF compilation are scriptable.

3) the data and experimental protocol justify the scope of the claim.

The current results support the first two conditions for score-level fusion but do not yet fully satisfy the third condition for naturally paired multimodal incidents. This limitation is explicit throughout the manuscript.

C. Contributions

This draft makes four scoped contributions:

- It defines VERA, a system-level anomaly-fusion pipeline with an explicit long-format multimodal schema.
- It defines RGA, a conservative reliability-aware fusion module that combines masked attention, calibration, score-distribution drift, sharpness, and a reliability gate.
- It adds a naturally paired MVTec 3D-AD RGB/3D benchmark path and builds RealFusion-LA, a reproducible label-aligned real-domain score-level stress benchmark using fraud, cyber, behavior, and text datasets.
- It reports baselines, missing-domain ablation, drift curves, adversarial score attacks, calibration, and counterfactual domain attribution with an explicit construct-validity boundary.

D. Evidence Boundary

The current quantitative evidence is strongest as a reliability-stress study over score-level fusion. It should not be read as a general claim that RGA is always superior to static attention, nor as evidence that the label-aligned benchmark captures naturally co-observed multimodal incidents. The paper therefore separates implemented artifacts, measured results, and remaining benchmark gaps.

Figure 1 shows the system boundary. Table I summarizes the implementation requirements used to keep the paper aligned with the codebase.

II. RELATED WORK

A. Anomaly Detection Systems

Deep anomaly detection has been studied across tabular, sequential, text, graph, and visual data. Pang et al. [3] emphasize that anomaly detection is tightly coupled to assumptions about labels, imbalance, and application context. VERA follows this systems view: it does not assume a single representation can absorb every domain cleanly. Instead, it treats each domain expert as a separate source of evidence.

B. Multimodal and Score-Level Fusion

Multimodal learning surveys distinguish early fusion, late fusion, hybrid fusion, and interaction-based fusion [2]. Early fusion concatenates features before prediction; late fusion combines model outputs; stacking learns a meta-model over domain predictions. Transformer attention [1] gives a natural mechanism for modeling interactions across available domains. The limitation is that ordinary attention learns a trust policy from training data and can remain overconfident when the inference condition changes.

C. RGB–3D Industrial Anomaly Detection

MVTec 3D-AD provides paired RGB and high-resolution 3D scans for industrial anomaly detection and localization [15]. Recent RGB–3D methods show that multimodal anomaly detection is not solved by direct feature concatenation alone. M3DM uses hybrid feature and decision-layer fusion with memory banks [16]; crossmodal feature mapping detects anomalies by checking consistency between RGB and point-cloud features [17]; Real3D-AD expands high-precision point-cloud benchmarking [18]; and noise-resistant multimodal work studies robustness when training data are not clean [19]. These papers motivate the paired benchmark path in VERA: the fusion layer should be judged on naturally paired modalities and on reliability stress, not only on clean score fusion.

D. Calibration, Drift, and Uncertainty

Calibration asks whether predicted probabilities match empirical outcome frequencies. Post-hoc calibration methods [4], [5], [6] improve probability quality, but calibration alone does not solve the problem of domain trust under drift. Uncertainty and calibration can degrade under dataset shift [7]. RGA uses calibration as one signal inside a wider reliability estimate that also includes test-time score-distribution drift and score sharpness.

E. Test-Time Adaptation

Test-time training and test-time adaptation update models at inference time to handle distribution shift [8], [9]. VERA now includes two score-level adaptation comparators: an entropy-minimizing Tent adapter over the fusion head and a high-confidence pseudo-label TTT adapter. They are intentionally conservative because the fusion setting receives domain scores rather than raw modality tensors. Their role is to test whether RGA adds value beyond generic score-level adaptation.

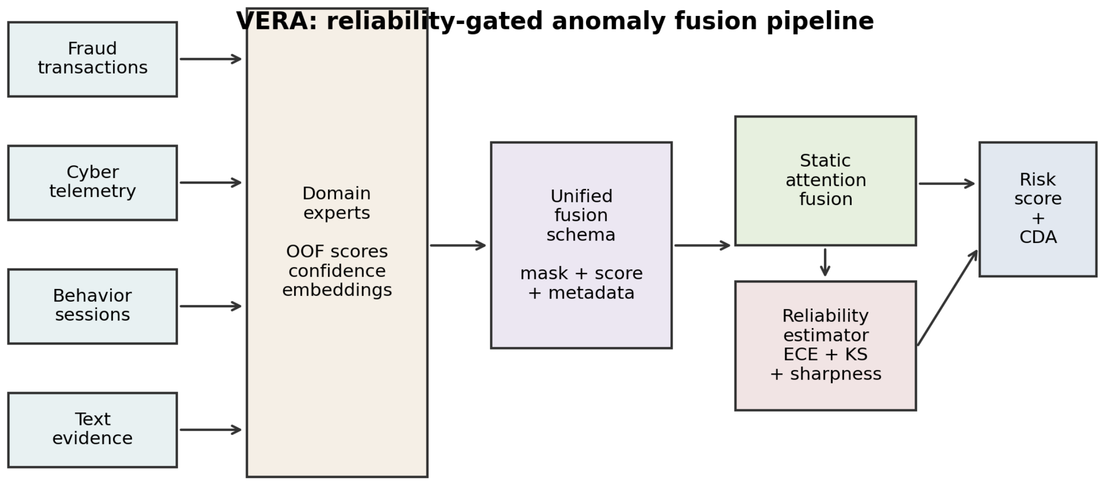

Fig. 1. VERA system architecture. Domain experts produce scores, confidence values, and compact embeddings; a unified fusion schema feeds static attention and the RGA reliability estimator; the system emits a risk score and counterfactual domain attribution.

F. Ensembles and Strong Baselines

Ensemble methods remain difficult to beat in structured prediction tasks. Random forests [10] and gradient boosting [11] are strong baselines for tabular and score-level settings. This is why the benchmark includes random forest fusion, early fusion MLP, late fusion ensemble, confidence-weighted mean, and static attention. A credible systems paper must compare against these practical alternatives, not only against weak averages.

G. Explanation for Fusion Systems

Feature attribution methods such as SHAP [12] explain individual model predictions, but fusion systems also need domain-level explanations. Analysts may need to know whether fraud evidence, network evidence, behavior evidence, or text evidence drove the risk score. VERA therefore includes counterfactual domain attribution: it masks one domain and reports the change in predicted risk.

III. System and Method

Each sample i contains up to D domain observations. Domain d contributes a feature vector $x_{i,d} \in \mathbb{R}^F$ containing a score, optional confidence, and compact embedding features. A binary mask $m_{i,d}$ indicates whether domain d is missing. The target label is $y_i \in \{0, 1\}$ when supervised labels are available.

The system estimates

$$\hat{p}_i = P(y_i = 1 \mid \{x_{i,d}, m_{i,d}\}_{d=1}^{D}), \tag{1}$$

where $\hat{p}_i$ is the final anomaly risk. In static fusion, the learned relationship among domains is fixed after training. In RGA, the system also computes a batch-level reliability vector $r \in [0, 1]^D$ at inference time and applies the sample mask to form effective per-sample weights $\tilde{r}_i \in [0, 1]^D$. The reliability vector should reduce trust in domains that appear miscalibrated, shifted, uninformative, or missing.

The evaluation problem is not only to maximize ROC-AUC. A useful verification system must also satisfy:

- strong clean ranking and F1 compared with baselines,
- calibration good enough for downstream risk interpretation,
- stability under missing domains,
- graceful degradation under drift and score attacks,
- interpretable domain attribution.

A. VERA System Architecture

1) Domain Experts: VERA assumes that each domain expert is trained or obtained separately. For the current benchmark, tabular scorers are trained for fraud, cyber, and behavior data, while a text scorer is trained for labeled text. The fusion layer does not require these experts to share the same raw feature space. It only requires score-level outputs and optional compact evidence features.

2) Fusion Schema: The unified schema is long-format: each row contains a sample identifier, a domain name, a binary label when available, a score, a confidence value, and embedding columns. This design allows different samples to have different available domains. Missing domains are represented by masks rather than by fabricated raw observations.

3) Static Attention Path: The static attention path encodes each available domain into a shared latent space and applies multi-head attention across domains. Domain embeddings allow the model to distinguish evidence sources. Missing domains are ignored through key-padding masks. This is the high-reliability default path.

4) Reliability-Aware Path: The reliability-aware path is activated when the average reliability of present domains falls below a configured threshold. This avoids the common failure mode where reliability weights disturb clean predictions even though no domain appears degraded. The reliability path is therefore a stress-response mechanism, not a universal replacement for the static model.

5) Explanation Path: For an analyst-facing explanation, the system masks one available domain at a time and recomputes risk. The resulting probability change gives a domain-level counterfactual attribution. This produces statements such as: if text evidence were absent, the risk would decrease by a given amount.

B. RGA Method

1) Domain Encoding and Attention: Each domain encoder maps the domain input vector into a shared latent space:

$$h_{i,d} = g_d(x_{i,d}) + e_d + p_d, \tag{2}$$

where e_d is a learned domain embedding and p_d is an optional learned position embedding used by the implementation. A multi-head self-attention block operates across available domain embeddings:

$$z_i = f_\theta(\mathrm{Pool}(\mathrm{Attention}(H_i, m_i))), \quad \hat{p}_i = \sigma(z_i). \tag{3}$$

2) Reliability Estimation: After model training, the system fits a reliability estimator on the validation split. For domain d, it stores validation score distributions, fits an isotonic calibrator when enough labels are available, and computes expected calibration error. At inference time:

$$r_d = \alpha(1 - \mathrm{ECE}_d) + \beta p_{\mathrm{KS},d} + \gamma S_d, \tag{4}$$

where ECE_d is validation calibration error, $p_{\mathrm{KS},d}$ is the Kolmogorov-Smirnov p-value comparing current scores with validation scores, and S_d is score sharpness from squared distance to 0.5. The current benchmark uses $(\alpha, \beta, \gamma) = (0.45, 0.35, 0.20)$.

3) Reliability Gate: Let $\bar{r}$ be the mean reliability over present domains and let $\tau = 0.66$. If $\bar{r} \geq \tau$, VERA uses the static attention path. If $\bar{r} < \tau$, reliability weights are injected into the fusion block:

$$\tilde{r}_{i,d} = (1 - m_{i,d}) r_{i,d}. \tag{5}$$

This makes RGA different from confidence weighting. Confidence weighting uses the local score confidence directly. RGA uses validation calibration, test-time drift evidence, and sharpness to decide when a domain should be trusted less.

4) Counterfactual Domain Attribution: For sample i, let $\hat{p}_i$ be the original prediction and $\hat{p}_{i,-d}$ be the prediction when domain d is masked. The attribution is

$$\Delta_{i,d} = \hat{p}_i - \hat{p}_{i,-d}. \tag{6}$$

A positive $\Delta_{i,d}$ indicates that domain d increased anomaly risk. A negative value indicates that domain d reduced risk.

TABLE II
RGA INFERENCE PROCEDURE.

Step	Operation
1	Encode all available domain vectors and construct the missing-domain mask.
2	Estimate domain reliability from validation ECE, KS score-drift evidence, and score sharpness.
3	Compute mean reliability over present domains.
4	If mean reliability is above threshold, use static attention inference.
5	Otherwise inject reliability weights into the attention fusion block.
6	Produce risk score and optional counterfactual domain attributions.

TABLE III
NATURALLY PAIRED MVTEC 3D-AD SUBSET METADATA.

Property	Value
Benchmark type	naturally_paired_mvtec3d_score_fusion
Natural pairing	yes
Categories	bagel
Samples	78
Positive fraction	0.410
Domains	rgb, depth_or_xyz

5) Algorithm Summary: The full inference procedure is summarized in Table II.

IV. BENCHMARK CONSTRUCTION

A. Naturally Paired MVTec 3D-AD Path

The primary benchmark path for this manuscript is MVTec 3D-AD. Each sample contains naturally co-observed RGB and 3D/depth evidence from the same object scan. The preparation script discovers paired files, emits two domain rows per sample, stores the original pairing key, and marks the metadata field `natural_pairing=true`. The initial scorer is deliberately lightweight: normal-reference image statistics produce reproducible RGB and depth anomaly scores without requiring a heavy specialized 3D detector. This keeps the first paired benchmark auditable; stronger RGB–3D scorers can replace these domain experts later while preserving the fusion interface.

Table III records the paired-subset metadata, and Table IV reports the paired smoke-run outcomes. The result is deliberately conservative: it proves that the RGB/3D path runs end to end, but it also shows that the current lightweight features and attention training are not yet competitive with tree and MLP fusion on this small subset.

Table V reports the metadata of the secondary label-aligned fusion input artifact.

B. Real-Domain Sources

RealFusion-LA is built from four local datasets: Credit Card Fraud, UNSW-NB15, Online Shoppers Intention, and labeled news text. Domain-specific out-of-fold scorers are trained first. Their scores, confidences, and compact embeddings are then converted into the common fusion schema.

TABLE IV
PAIRED MVTEC 3D-AD BAGEL-SUBSET SMOKE BENCHMARK. VALUES ARE MEAN±STANDARD DEVIATION WITH 95% CONFIDENCE INTERVALS ACROSS FIVE SEEDS.

Method	ROC-AUC	PR-AUC	F1	ECE
Random forest	**0.9270±0.0940 [0.7651, 1.0000]**	**0.9073±0.1240 [0.6928, 1.0000]**	**0.8960±0.0740 [0.8024, 0.9933]**	0.2151±0.0264 [0.1786, 0.2514]
Conf.-weighted mean	0.8730±0.1100 [0.6857, 0.9667]	0.8831±0.1040 [0.7027, 0.9613]	0.6758±0.1614 [0.4000, 0.8227]	0.2144±0.0496 [0.1650, 0.2840]
Tent score adapter	0.9048±0.0858 [0.7619, 0.9952]	0.8919±0.1030 [0.7172, 0.9938]	0.6781±0.0949 [0.5167, 0.7650]	0.2088±0.0605 [0.1480, 0.3037]
TTT pseudo-label	0.8825±0.0918 [0.7413, 0.9937]	0.8850±0.0988 [0.7222, 0.9937]	0.8186±0.1098 [0.6769, 0.9857]	0.1706±0.0752 [0.0814, 0.2907]
Early fusion MLP	0.9175±0.0991 [0.7492, 1.0000]	0.8965±0.1281 [0.6776, 1.0000]	0.8545±0.1014 [0.7206, 0.9923]	0.1807±0.0559 [0.1379, 0.2780]
Late fusion ensemble	0.9016±0.0911 [0.7476, 0.9937]	0.8783±0.1186 [0.6764, 0.9933]	0.7615±0.0980 [0.5912, 0.8333]	0.2417±0.0424 [0.1837, 0.3027]
Static attention	0.4952±0.1514 [0.2683, 0.6619]	0.5374±0.1078 [0.3737, 0.6514]	0.1800±0.1568 [0.0000, 0.3850]	**0.1299±0.0359 [0.0751, 0.1716]**
RGA attention	0.4952±0.1514 [0.2683, 0.6619]	0.5374±0.1078 [0.3737, 0.6514]	0.1800±0.1568 [0.0000, 0.3850]	**0.1299±0.0359 [0.0751, 0.1716]**

TABLE V
BENCHMARK METADATA FOR THE SECONDARY LABEL-ALIGNED FUSION INPUT ARTIFACT.

Property	Value
Benchmark type	label_aligned_real_domain_score_fusion
Natural pairing	no
Pairing unit	binary-label-aligned composite sample
Samples	8000
Positive fraction	0.307
Scorer train fraction	–
Domains	fraud, cyber, behavior, nlp

TABLE VI
OUT-OF-FOLD DOMAIN SCORER QUALITY USED TO CONSTRUCT REALFUSION-LA.

Domain	Rows	Pos. rate	OOF ROC	OOF PR
fraud	20000	0.025	0.9686	0.7826
cyber	20000	0.500	0.9684	0.9696
behavior	12330	0.155	0.8716	0.6053
nlp	20000	0.500	0.9973	0.9973

The generated fusion table contains 8,000 composite samples, four domains, eight embedding features per domain, a 0.307 positive rate, and 11.8% nominal missingness. Domain coverage is 0.883 for fraud, 0.885 for cyber, 0.885 for behavior, and 0.877 for text.

C. Label Alignment Limitation

The datasets do not share natural entity identifiers or timestamps. Therefore, composite samples are label-aligned: a positive composite draws positive records from available domains, and a negative composite draws negative records. This is a meaningful real-domain score-fusion benchmark, but it is not evidence that the system has learned naturally paired multimodal incident structure. That distinction is central to the claim boundary of the paper. The metadata now records this explicitly with `natural_pairing=false` for RealFusion-LA.

D. Training Protocol

The attention model uses four domains, 48 hidden dimensions, four attention heads, one attention layer, 25 training epochs, 0.15 domain dropout during training, and seeds 42 through 46. Clean benchmark results are reported as mean±standard deviation across five seeds. The experiment runner also repeats missing-domain, drift, adversarial, calibration, and CDA analyses for every configured seed, then stores both raw per-seed rows and aggregated mean, standard deviation, and 95% confidence intervals.

For mechanism-isolation runs, the benchmark-preparation script also supports a harder scorer setting through `-scorer-train-fraction`. Values such as 0.05 or 0.10 train each out-of-fold domain scorer on only a stratified fraction of its training fold before fusion rows are built. This is intended to reduce clean-score saturation and expose whether the fusion mechanism helps when domain experts are useful but imperfect.

E. Baselines and Metrics

Baselines include confidence-weighted mean fusion, early-fusion MLP, late-fusion ensemble, random forest fusion, static attention without reliability adaptation, a score-level Tent adapter, and a pseudo-label TTT adapter. Metrics include ROC-AUC, PR-AUC, F1, expected calibration error, Brier score, and accuracy. Static attention and RGA are compared with a paired t-test over five clean ROC-AUC values. ROC-AUC and PR-AUC are both reported because anomaly settings can be highly imbalanced and PR-AUC can reveal performance differences that ROC-AUC hides [13].

F. Confidence Intervals

Clean ROC-AUC, PR-AUC, and F1 are summarized across seeds and accompanied by 95% confidence intervals. Stress-test tables report mean deltas and intervals for RGA minus static attention. The result JSON stores raw per-seed stress rows so that the interval computation is auditable rather than embedded only in the manuscript.

G. Secondary Clean Sanity Check

Table VII and Fig. 2 show clean held-out performance. This table is intentionally treated as a sanity check, not as the headline result, because the label-aligned split is near saturated. Bold values indicate the highest rounded mean in each column; multiple bold values are rounded ties.

Seed-level confidence intervals for the same clean split are reported in Table VIII.

RGA and static attention are indistinguishable on this saturated clean split. The clean benchmark should therefore be

TABLE VII

SATURATED CLEAN HELD-OUT PERFORMANCE ON REALFUSION-LA. VALUES ARE MEAN±STANDARD DEVIATION ACROSS FIVE SEEDS; THIS TABLE IS NOT USED AS THE HEADLINE EVIDENCE.

Method	ROC-AUC	PR-AUC	F1	ECE	Brier
Random forest	0.9997±0.0002 [0.9994, 0.9999]	0.9994±0.0004 [0.9988, 0.9998]	**0.9906±0.0008 [0.9898, 0.9918]**	0.0287±0.0007 [0.0280, 0.0298]	0.0076±0.0006 [0.0067, 0.0083]
Conf.-weighted mean	0.9982±0.0007 [0.9973, 0.9993]	0.9955±0.0027 [0.9914, 0.9986]	0.9794±0.0042 [0.9739, 0.9837]	0.0834±0.0013 [0.0816, 0.0849]	0.0222±0.0011 [0.0211, 0.0241]
Tent score adapter	**0.9997±0.0001 [0.9995, 0.9998]**	0.9994±0.0002 [0.9990, 0.9996]	0.9868±0.0017 [0.9840, 0.9887]	0.0062±0.0009 [0.0049, 0.0073]	0.0065±0.0009 [0.0050, 0.0076]
TTT pseudo-label	0.9997±0.0001 [0.9995, 0.9998]	0.9993±0.0003 [0.9988, 0.9996]	0.9853±0.0024 [0.9812, 0.9877]	0.0045±0.0011 [0.0030, 0.0061]	0.0062±0.0010 [0.0048, 0.0074]
Early fusion MLP	**0.9997±0.0001 [0.9996, 0.9998]**	0.9993±0.0002 [0.9992, 0.9995]	0.9837±0.0043 [0.9780, 0.9877]	0.0073±0.0027 [0.0045, 0.0113]	0.0072±0.0015 [0.0051, 0.0089]
Late fusion ensemble	0.9996±0.0001 [0.9994, 0.9998]	0.9992±0.0002 [0.9988, 0.9995]	0.9832±0.0029 [0.9790, 0.9858]	0.0103±0.0010 [0.0092, 0.0116]	0.0073±0.0007 [0.0063, 0.0081]
Static attention	**0.9997±0.0001 [0.9996, 0.9999]**	**0.9994±0.0002 [0.9992, 0.9997]**	0.9896±0.0012 [0.9879, 0.9908]	**0.0041±0.0005 [0.0034, 0.0049]**	**0.0050±0.0005 [0.0043, 0.0055]**
RGA attention	**0.9997±0.0001 [0.9996, 0.9999]**	**0.9994±0.0002 [0.9992, 0.9998]**	0.9894±0.0010 [0.9879, 0.9907]	0.0042±0.0004 [0.0038, 0.0049]	**0.0051±0.0004 [0.0044, 0.0055]**

TABLE VIII

CLEAN RANKING AND F1 CONFIDENCE INTERVALS ACROSS CONFIGURED SEEDS.

Method	ROC-AUC 95% CI	PR-AUC 95% CI	F1 95% CI
Random forest	[0.9994, 0.9999]	[0.9988, 0.9998]	[0.9898, 0.9918]
Conf.-weighted mean	[0.9973, 0.9993]	[0.9914, 0.9986]	[0.9739, 0.9837]
Tent score adapter	[0.9995, 0.9998]	[0.9990, 0.9996]	[0.9840, 0.9887]
TTT pseudo-label	[0.9995, 0.9998]	[0.9988, 0.9996]	[0.9812, 0.9877]
Early fusion MLP	[0.9996, 0.9998]	[0.9992, 0.9995]	[0.9780, 0.9877]
Late fusion ensemble	[0.9994, 0.9998]	[0.9988, 0.9995]	[0.9790, 0.9858]
Static attention	[0.9996, 0.9999]	[0.9992, 0.9997]	[0.9879, 0.9908]
RGA attention	[0.9996, 0.9999]	[0.9992, 0.9998]	[0.9879, 0.9907]

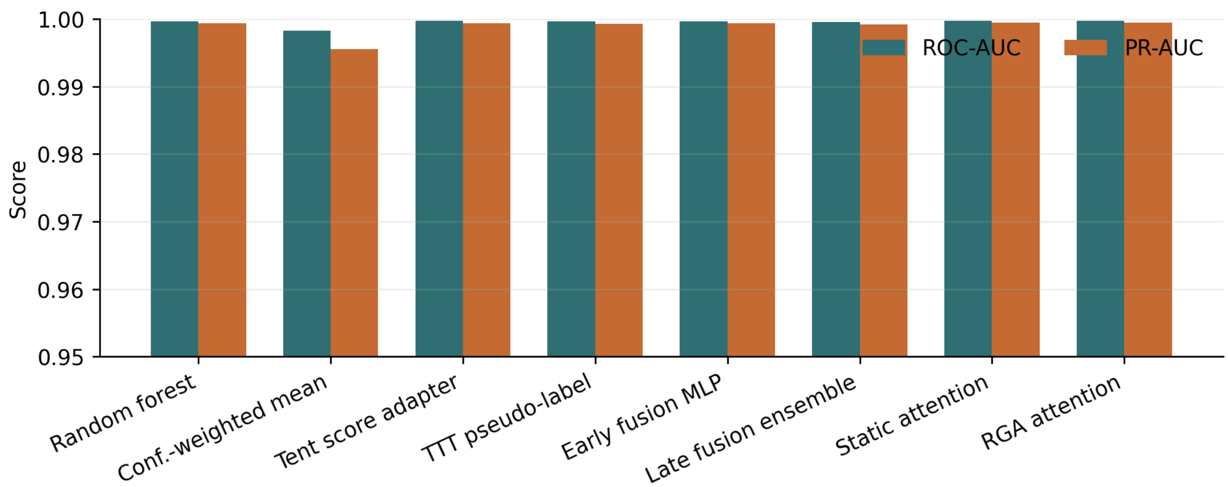

Fig. 2. Clean benchmark ROC-AUC and PR-AUC across score-level and attention fusion methods.

interpreted as competitive preservation of performance, not as evidence of a clean-split improvement.

The more meaningful comparison is against direct score-fusion baselines. RGA improves over confidence-weighted mean fusion by about 0.0015 ROC-AUC, 0.0039 PR-AUC, 0.0100 F1, and reduces ECE from 0.0834 to 0.0042. Random forest fusion obtains the best F1 at 0.9906, but its ECE is 0.0287, substantially worse than attention-based fusion. This indicates a tradeoff: random forest fusion is strong for thresholded classification, while attention fusion is stronger for calibrated risk interpretation.

V. ROBUSTNESS AND ABLATION RESULTS

A. *Missing-Domain Ablation*

Table IX and Fig. 3 report additional inference-time domain dropout. At zero extra dropout, RGA is equal to static attention, and the same equality holds through the configured dropout sweep. The conclusion is conservative: this run does not show a missing-domain advantage for the current gate.

B. *Score Drift*

Table X summarizes degradation area under per-domain score-noise sweeps. Higher area means slower degradation. All deltas round to zero or near zero because clean performance is near saturated, so this table is best read as a robustness check rather than a win. The full curves are shown in Fig. 4.

TABLE IX
MISSING-DOMAIN ABLATION. DELTA IS RGA MINUS STATIC ATTENTION.

Dropout	Static ROC	RGA ROC	Delta	Delta 95% CI
0.0	0.9997	0.9997	0.0000	[0.0000, 0.0000]
0.1	0.9992	0.9992	0.0000	[0.0000, 0.0000]
0.2	0.9980	0.9980	0.0000	[0.0000, 0.0000]
0.3	0.9959	0.9959	0.0000	[0.0000, 0.0000]
0.5	0.9786	0.9786	0.0000	[0.0000, 0.0000]

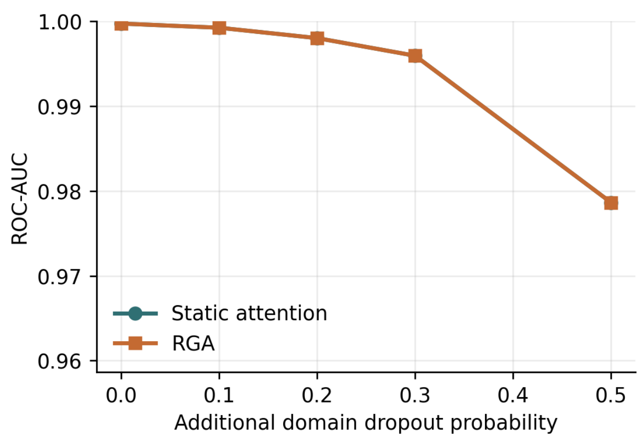

Fig. 3. ROC-AUC under additional missing-domain dropout.

C. Adversarial Score Perturbations

Table XI evaluates all score-level perturbations, while Fig. 5 isolates the all-domain attacks so the meaningful deltas are visible. Single-domain attacks are mostly ties. The important stress case is coherent all-domain score collapse. Under the all-domain zero attack, RGA improves ROC-AUC from 0.7130 to 0.7770. Under the all-domain max attack, it improves ROC-AUC from 0.7114 to 0.7418. Under all-domain Gaussian noise, it is slightly worse by 0.0003 ROC-AUC.

D. Mechanism Isolation Protocol

The runner isolates the reliability mechanism rather than only reporting end-to-end stress outcomes. It emits two diagnostics when the mechanism-isolation block is enabled. First, a τ sweep evaluates $\tau \in \{0.4, 0.5, 0.6, 0.66, 0.7, 0.8, 0.9\}$ on clean data and all-domain stress conditions, recording both ROC-AUC deltas and the adaptation rate. Second, a reliability-component ablation compares the full estimator with no-ECE, no-KS, no-sharpness, and no-gate variants on all-domain score attacks. These diagnostics show whether the gate and reliability terms are active contributors rather than incidental implementation details.

Threshold-sweep outcomes and adaptation rates are reported in Table XII.

At the configured $\tau = 0.66$, clean adaptation is rare: the mean adaptation rate is 0.032 with a confidence interval that includes zero. This explains why the missing-domain deltas

TABLE X
DRIFT ROBUSTNESS SUMMARY. HIGHER AREA IS BETTER.

Domain	Static	RGA	Delta	Seeds
behavior	0.2999 [0.2999, 0.3000]	0.2999 [0.2999, 0.3000]	-0.0000	5
cyber	0.2999 [0.2999, 0.3000]	0.2999 [0.2998, 0.3000]	-0.0000	5
fraud	0.2999 [0.2999, 0.3000]	0.2999 [0.2999, 0.3000]	0.0000	5
nlp	0.2998±0.0001 [0.2998, 0.2999]	0.2998±0.0001 [0.2998, 0.2999]	-0.0000	5

are effectively zero: the gate usually keeps the static path active. Under all-domain zero and max score collapse, the same threshold activates the reliability path for every evaluated sample and yields the measured gains. Lower thresholds through $\tau = 0.60$ suppress adaptation and lose those gains; higher thresholds adapt on clean data more often and slightly hurt clean ranking.

Per-component contributions on all-domain score attacks are reported in Table XIII.

The component ablation produces a useful negative result. Removing the ECE term does not hurt the all-domain collapse cases; it improves the max-attack delta from 0.0304 to 0.0428 and the zero-attack delta from 0.0640 to 0.0691. In contrast, removing the KS drift term eliminates adaptation and returns exactly to static attention. This suggests that, in the current stress benchmark, the distribution-shift signal is the necessary trigger while validation ECE is not the driver of the robustness gain.

VI. CALIBRATION AND EXPLANATION

A. Calibration

Table XIV reports calibration for the final configured seed. Static attention is slightly better than RGA on ECE and Brier score in that seed. Both attention variants remain much better calibrated than confidence-weighted mean fusion in the five-seed clean table. The reliability diagram is shown in Fig. 6.

B. Counterfactual Domain Attribution

Counterfactual domain attribution was computed for 400 test samples across configured seeds. Mean absolute probability changes were 0.0138 for fraud, 0.0172 for cyber, 0.0059 for behavior, and 0.0353 for text. Text has the largest average counterfactual impact, consistent with the strong text scorer in Table VI; the per-domain magnitudes are visualized in Fig. 7.

VII. DISCUSSION AND LIMITATIONS

A. What the Results Support

The results support three concrete claims. First, RGA preserves clean ranking performance relative to static attention. Second, RGA is far better than confidence-weighted mean fusion on this benchmark, especially for calibration. Third, reliability adaptation improves the severe all-domain zero/max score-collapse stress cases. These are useful systems results, but they are bounded by the benchmark design.

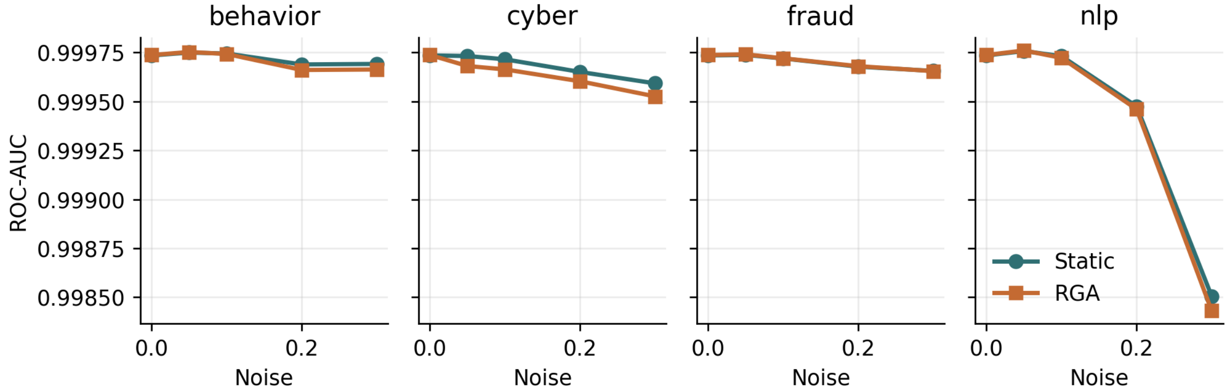

Fig. 4. Per-domain score-drift curves on the real-domain benchmark.

TABLE XI
ADVERSARIAL PERTURBATION BENCHMARK. DELTA IS RGA MINUS STATIC ATTENTION.

Attack	Target	Static ROC	RGA ROC	Delta	Delta 95% CI
gaussian_noise	all	0.9997	0.9994	-0.0003	[-0.0005, -0.0001]
gaussian_noise	behavior	0.9997	0.9997	-0.0000	[-0.0001, 0.0000]
gaussian_noise	cyber	0.9997	0.9997	-0.0001	[-0.0001, -0.0000]
gaussian_noise	fraud	0.9997	0.9997	0.0000	[0.0000, 0.0000]
gaussian_noise	nlp	0.9997	0.9997	-0.0000	[-0.0001, 0.0000]
max_attack	all	0.7114	0.7418	0.0304	[0.0172, 0.0433]
max_attack	behavior	0.9996	0.9996	0.0000	[0.0000, 0.0000]
max_attack	cyber	0.9960	0.9961	0.0000	[-0.0000, 0.0001]
max_attack	fraud	0.9987	0.9987	0.0000	[-0.0000, 0.0000]
max_attack	nlp	0.9735	0.9735	0.0000	[0.0000, 0.0000]
zero_attack	all	0.7130	0.7770	0.0640	[0.0447, 0.0975]
zero_attack	behavior	0.9995	0.9995	0.0000	[0.0000, 0.0001]
zero_attack	cyber	0.9967	0.9967	0.0000	[-0.0000, 0.0001]
zero_attack	fraud	0.9989	0.9989	-0.0000	[-0.0000, 0.0000]
zero_attack	nlp	0.9827	0.9827	0.0000	[0.0000, 0.0000]

B. What the Results Do Not Support

The results do not support a broad claim that RGA is always better than static attention. The clean ROC-AUC difference is not significant, missing-domain benefits are limited, and drift results are mixed. The benchmark also does not prove true multimodal incident reconstruction because the domains are label-aligned rather than naturally co-observed.

C. Failure Analysis

The result artifact exports representative cases for the largest RGA win, largest RGA loss, cases where RGA fixes static attention, cases where static attention fixes RGA, and high-confidence RGA failures. Each case includes the sample identifier, label, static and RGA probabilities, per-domain scores, missing-domain mask, and reliability weights.

Representative cases are listed in Table XV.

The current benchmark exposes four important failure modes:

- the label-aligned construction makes the clean split highly separable,
- the conservative reliability gate suppresses adaptation under moderate missingness,
- random forest fusion remains the strongest F1 baseline,
- the naturally paired path is still a small smoke benchmark rather than full-dataset evidence.

TABLE XII
RELIABILITY-GATE THRESHOLD SWEEP. DELTA IS RGA MINUS STATIC ATTENTION; ADAPTATION RATE IS THE FRACTION OF EVALUATED SAMPLES USING THE RELIABILITY-AWARE PATH.

Condition	τ	Static ROC	RGA ROC	Delta	Delta 95% CI	Adapt rate
clean	0.40	0.9997	0.9997	0.0000	[0.0000, 0.0000]	0.000
clean	0.50	0.9997	0.9997	0.0000	[0.0000, 0.0000]	0.000
clean	0.60	0.9997	0.9997	0.0000	[0.0000, 0.0000]	0.000
clean	0.66	0.9997	0.9997	0.0000	[0.0000, 0.0000]	0.032
clean	0.70	0.9997	0.9997	-0.0000	[-0.0001, 0.0000]	0.128
clean	0.80	0.9997	0.9996	-0.0001	[-0.0003, -0.0001]	0.672
clean	0.90	0.9997	0.9995	-0.0003	[-0.0008, -0.0001]	1.000
gaussian_noise:all	0.40	0.9997	0.9997	0.0000	[0.0000, 0.0000]	0.000
gaussian_noise:all	0.50	0.9997	0.9997	0.0000	[0.0000, 0.0000]	0.000
gaussian_noise:all	0.60	0.9997	0.9996	-0.0001	[-0.0003, -0.0001]	0.672
gaussian_noise:all	0.66	0.9997	0.9995	-0.0002	[-0.0004, -0.0001]	1.000
gaussian_noise:all	0.70	0.9997	0.9995	-0.0002	[-0.0004, -0.0001]	1.000
gaussian_noise:all	0.80	0.9997	0.9995	-0.0002	[-0.0004, -0.0001]	1.000
gaussian_noise:all	0.90	0.9997	0.9995	-0.0002	[-0.0004, -0.0001]	1.000
max_attack:all	0.40	0.7114	0.7114	0.0000	[0.0000, 0.0000]	0.000
max_attack:all	0.50	0.7114	0.7114	0.0000	[0.0000, 0.0000]	0.000
max_attack:all	0.60	0.7114	0.7114	0.0000	[0.0000, 0.0000]	0.000
max_attack:all	0.66	0.7114	0.7418	0.0304	[0.0172, 0.0433]	1.000
max_attack:all	0.70	0.7114	0.7418	0.0304	[0.0172, 0.0433]	1.000
max_attack:all	0.80	0.7114	0.7418	0.0304	[0.0172, 0.0433]	1.000
max_attack:all	0.90	0.7114	0.7418	0.0304	[0.0172, 0.0433]	1.000
zero_attack:all	0.40	0.7130	0.7130	0.0000	[0.0000, 0.0000]	0.000
zero_attack:all	0.50	0.7130	0.7130	0.0000	[0.0000, 0.0000]	0.000
zero_attack:all	0.60	0.7130	0.7130	0.0000	[0.0000, 0.0000]	0.000
zero_attack:all	0.66	0.7130	0.7770	0.0640	[0.0447, 0.0975]	1.000
zero_attack:all	0.70	0.7130	0.7770	0.0640	[0.0447, 0.0975]	1.000
zero_attack:all	0.80	0.7130	0.7770	0.0640	[0.0447, 0.0975]	1.000
zero_attack:all	0.90	0.7130	0.7770	0.0640	[0.0447, 0.0975]	1.000

D. Internal Validity

The benchmark scripts are deterministic for configured seeds, and clean results are reported across five seeds. The runner repeats robustness ablations across configured seeds and stores confidence intervals. Residual risk remains if the stress distributions are too synthetic relative to deployment failures.

E. Construct Validity

RealFusion-LA measures real-domain score fusion, not naturally paired multimodal incident detection. The MVTec 3D-AD path addresses natural RGB/3D pairing, but the current local run is a small bagel-subset smoke benchmark with lightweight features. This remains the largest construct-validity threat. The paper therefore avoids claiming that the model has learned causal or temporal relationships among fraud, cyber, behavior, and text domains, or that the paired path is competitive with specialized RGB–3D anomaly detectors.

F. External Validity

The datasets are useful but not sufficient for deployment claims. A production system would need domain-specific label definitions, temporal splits, deployment-time calibration monitoring, privacy review, and dataset license review.

G. Statistical Validity

Five clean seeds provide only limited statistical support, especially on a near-saturated split. This draft therefore reports seed-level confidence intervals for clean PR-AUC and F1 and treats stress-test deltas as the primary diagnostic. DeLong tests are still used where assumptions hold [14], but the paper avoids treating a single significance test as proof of broad superiority.

VIII. REPRODUCIBILITY

The reproducibility flow has three scripted steps:

1) build the real-domain fusion input table,

TABLE XIII
RELIABILITY-COMPONENT ABLATION ON ALL-DOMAIN SCORE ATTACKS. DELTA IS RGA MINUS STATIC ATTENTION.

Variant	Attack	Static ROC	RGA ROC	Delta	Delta 95% CI	Adapt rate
full	gaussian_noise	0.9997	0.9994	-0.0003	[-0.0005, -0.0001]	0.992
full	max_attack	0.7114	0.7418	0.0304	[0.0172, 0.0433]	1.000
full	zero_attack	0.7130	0.7770	0.0640	[0.0447, 0.0975]	1.000
no_ece	gaussian_noise	0.9997	0.9992	-0.0005	[-0.0008, -0.0002]	1.000
no_ece	max_attack	0.7114	0.7542	0.0428	[0.0067, 0.0790]	1.000
no_ece	zero_attack	0.7130	0.7821	0.0691	[0.0554, 0.0911]	1.000
no_gate	gaussian_noise	0.9997	0.9994	-0.0003	[-0.0005, -0.0001]	1.000
no_gate	max_attack	0.7114	0.7418	0.0304	[0.0172, 0.0433]	1.000
no_gate	zero_attack	0.7130	0.7770	0.0640	[0.0447, 0.0975]	1.000
no_ks	gaussian_noise	0.9997	0.9997	0.0000	[0.0000, 0.0000]	0.000
no_ks	max_attack	0.7114	0.7114	0.0000	[0.0000, 0.0000]	0.000
no_ks	zero_attack	0.7130	0.7130	0.0000	[0.0000, 0.0000]	0.000
no_sharpness	gaussian_noise	0.9997	0.9994	-0.0004	[-0.0006, -0.0001]	1.000
no_sharpness	max_attack	0.7114	0.7474	0.0360	[0.0170, 0.0572]	1.000
no_sharpness	zero_attack	0.7130	0.7789	0.0659	[0.0477, 0.0980]	1.000

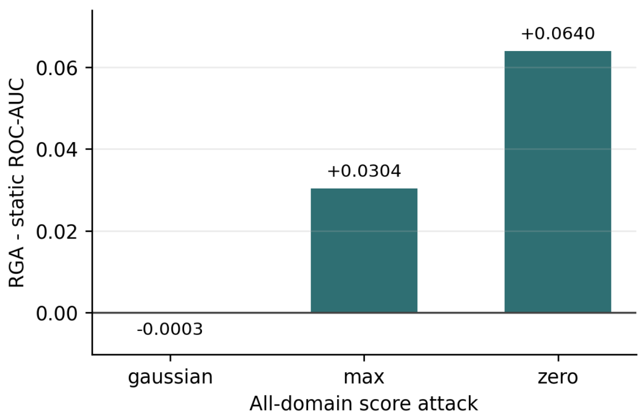

Fig. 5. All-domain adversarial ROC-AUC delta between RGA and static attention. Complete single-domain scenarios are reported in Table XI.

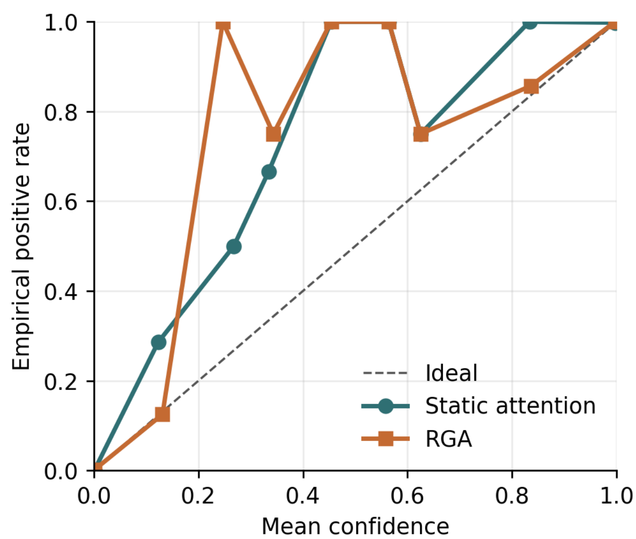

Fig. 6. Reliability diagram for static attention and RGA.

TABLE XIV
CALIBRATION AND COUNTERFACTUAL DOMAIN-ATTRIBUTION SUMMARY.

Metric	Static	RGA
ECE	0.0041	0.0042
Brier	0.0050	0.0051
CDA samples	–	400
CDA/ECE Spearman	–	–

TABLE XV
REPRESENTATIVE FAILURE AND CONTRAST CASES FROM THE LATEST CONFIGURED SEED.

Case	Label	Static p	RGA p	Sample
biggest_rga_win	0	0.2888	0.1359	real_fusion_007461
biggest_rga_loss	1	0.9521	0.3696	real_fusion_004504
high_confidence_rga_failure	1	0.1083	0.0620	real_fusion_003452

2) run the real fusion benchmark with the YAML configuration,
3) generate tables, figures, and the PDF manuscript.

The implementation uses the local scripts under `src/scripts/`, the configurations `configs/attention_real_fusion.yaml` and `configs/attention_mvtec3d_fusion.yaml`, result artifacts under `experiments/fusion/`, and generated paper assets under `docs/research/`. The paired benchmark entry point is `src/scripts/prepare_`

Fig. 7. Mean absolute counterfactual domain-attribution impact.

`mvtec3d_fusion_benchmark.py`. Targeted regression tests cover attention masking, statistics, baselines, reliability estimation, counterfactual explanation, paired-benchmark metadata, and result aggregation.

IX. Conclusion

VERA presents a reliability-aware anomaly-fusion pipeline with domain experts, unified score schema, masked attention, reliability-gated adaptation, counterfactual explanation, benchmark construction, baselines, ablations, calibration, and failure analysis. The strongest measured claim is that RGA preserves clean attention performance, improves over confidence-weighted mean fusion, and helps under severe all-domain score collapse. The main remaining research gap is full naturally paired multimodal validation with stronger RGB–3D domain experts.

References

[1] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, "Attention is all you need," in *Advances in Neural Information Processing Systems*, 2017.

[2] T. Baltrusaitis, C. Ahuja, and L.-P. Morency, "Multimodal machine learning: A survey and taxonomy," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 41, no. 2, pp. 423–443, 2019.

[3] G. Pang, C. Shen, L. Cao, and A. van den Hengel, "Deep learning for anomaly detection: A review," *ACM Computing Surveys*, vol. 54, no. 2, 2021.

[4] B. Zadrozny and C. Elkan, "Transforming classifier scores into accurate multiclass probability estimates," in *Proceedings of the ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2002.

[5] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in *Proceedings of the International Conference on Machine Learning*, 2017.

[6] J. Platt, "Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods," in *Advances in Large Margin Classifiers*, 1999.

[7] Y. Ovadia et al., "Can you trust your model's uncertainty? Evaluating predictive uncertainty under dataset shift," in *Advances in Neural Information Processing Systems*, 2019.

[8] Y. Sun, X. Wang, Z. Liu, J. Miller, A. Efros, and M. Hardt, "Test-time training with self-supervision for generalization under distribution shifts," in *Proceedings of the International Conference on Machine Learning*, 2020.

[9] D. Wang, E. Shelhamer, S. Liu, B. Olshausen, and T. Darrell, "Tent: Fully test-time adaptation by entropy minimization," in *International Conference on Learning Representations*, 2021.

[10] L. Breiman, "Random forests," *Machine Learning*, vol. 45, pp. 5–32, 2001.

[11] J. H. Friedman, "Greedy function approximation: A gradient boosting machine," *The Annals of Statistics*, vol. 29, no. 5, pp. 1189–1232, 2001.

[12] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Advances in Neural Information Processing Systems*, 2017.

[13] J. Davis and M. Goadrich, "The relationship between precision-recall and ROC curves," in *Proceedings of the International Conference on Machine Learning*, 2006.

[14] E. R. DeLong, D. M. DeLong, and D. L. Clarke-Pearson, "Comparing the areas under two or more correlated receiver operating characteristic curves: A nonparametric approach," *Biometrics*, vol. 44, no. 3, pp. 837–845, 1988.

[15] P. Bergmann, X. Jin, D. Sattlegger, and C. Steger, "The MVTec 3D-AD Dataset for Unsupervised 3D Anomaly Detection and Localization," in *Proceedings of VISAPP*, 2022.

[16] Y. Wang, J. Peng, J. Zhang, R. Yi, Y. Wang, and C. Wang, "Multimodal Industrial Anomaly Detection via Hybrid Fusion," in *Proceedings of CVPR*, 2023.

[17] A. Costanzino, P. Zama Ramirez, G. Lisanti, and L. Di Stefano, "Multimodal Industrial Anomaly Detection by Crossmodal Feature Mapping," *arXiv preprint arXiv:2312.04521*, 2023.

[18] J. Liu, G. Xie, R. Chen, X. Li, J. Wang, Y. Liu, C. Wang, and F. Zheng, "Real3D-AD: A Dataset of Point Cloud Anomaly Detection," *arXiv preprint arXiv:2309.13226*, 2023.

[19] C. Wang, H. Zhu, J. Peng, Y. Wang, R. Yi, Y. Wu, L. Ma, and J. Zhang, "M3DM-NR: RGB-3D Noisy-Resistant Industrial Anomaly Detection via Multimodal Denoising," *arXiv preprint arXiv:2406.02263*, 2024.